

Closed Affine Flats with an Empty Minsum Locus

A positive-distance counterexample and an exact Hilbert-space criterion

Candidate result for arXiv:1410.3690

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Status: candidate counterexample, likely valid. The proof is complete and self-contained; novelty remains subject to human literature review.

1 Source question and precise scope

Jahn, Kupitz, Martini, and Richter study finite-dimensional generalized Minkowski spaces and finite families of convex target sets [1]. Their Section 6 asks what survives in infinite dimension (source PDF p. 34):

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6 Perspectives and Interesting Open Problems

It is clear that one can go on with generalizing minsum location problems based on our results in a very natural way. We list now different possibilities for further research in such directions. As the reader will observe, there are various natural overlaps between these problems and topics.

We have considered the case where the dimension of X is finite. This way we could use arguments of compactness, and there was only one vector space topology on X . But what happens if the dimension of X is infinite? What concepts and statements are preserved, and what is different? And what are appropriate additional assumptions to obtain results similar to the finite-dimensional setting?

Exact transcription. “We have considered the case where the dimension of X is finite. This way we could use arguments of compactness, and there was only one vector space topology on X . But what happens if the dimension of X is infinite? What concepts and statements are preserved, and what is different? And what are appropriate additional assumptions to obtain results similar to the finite-dimensional setting?”

The finite-dimensional statement tested here is Proposition 4.1(c): for affine-flat targets, the Fermat–Torricelli locus is nonempty. The source’s definition and proposition appear on PDF p. 21:

$$\inf_{x \in X} \sum_{i=1}^n \text{dist}_{\gamma_i}(x, K_i). \quad (9)$$

We start the discussion of (9) with an analogue of Proposition 3.1. As above, we call the set of all minimizers of (9) the *Fermat–Torricelli locus* of (9). A particular emphasis is on the case where all the sets K_i are affine flats (i.e., affine subspaces) in X .

Proposition 4.1 (a) *The Fermat–Torricelli locus of (9) is closed and convex.*

(b) *If one of the sets K_i , $i \in \{1, \dots, n\}$, is bounded, then the Fermat–Torricelli locus of (9) is bounded and non-empty.*

(c) *If all the sets K_i , $i \in \{1, \dots, n\}$, are affine flats, then the Fermat–Torricelli locus of (9) is non-empty. Moreover, it is a Minkowski sum of a closed, bounded, convex set and a linear space that may be degenerated to $\{0\}$.*

We give a full negative answer to the natural infinite-dimensional extension of this statement, already for two closed affine flats and the ordinary Hilbert norm. We also determine exactly when the two-flat conclusion does survive.

2 Definitions and proof intuition

For a nonempty subset C of a real Hilbert space $\mathcal{H}$, write

$$d(x, C) = \inf_{c \in C} \|x - c\|.$$

For two target sets A, B , their minsum objective and locus are

$$F(x) = d(x, A) + d(x, B), \quad \text{argmin } F.$$

A closed affine flat has the form $a + U$, where U is a closed linear subspace.

Proof Intuition

The finite-dimensional proof can force a minimizing sequence to converge because sums of linear subspaces are closed. Infinite-dimensional Hilbert spaces have a precise new obstruction: two closed subspaces can have a nonclosed algebraic sum. For $A = a + U$ and $B = b + V$, finding a nearest pair is exactly the problem of best-approximating $b - a$ by $U + V$. The orthogonal projection onto $\overline{U + V}$ always exists, but it need not lie in $U + V$. That single missing membership is equivalent to an empty minsum locus.

For the explicit example, take the graph of the compact diagonal operator $D(x)_n = x_n/n$. Its range is dense and nonclosed. The vector $y = (1/n)$ is square summable but has only the formal preimage $(1, 1, \dots)$, which is not in ℓ^2 . Adding one orthogonal real coordinate makes the unattained distance strictly positive, namely one.

3 Exact theorem for two affine flats

Theorem 1 (Two-flat criterion and locus formula). *Let $\mathcal{H}$ be a real Hilbert space, let $A = a + U$ and $B = b + V$ be nonempty closed affine flats, and put*

$$w = b - a, \quad L = U + V, \quad P = P_{\overline{L}}.$$

Then:

(i)

$$\inf_{x \in \mathcal{H}} (d(x, A) + d(x, B)) = \text{dist}(A, B) = \text{dist}(w, L) = \|(I - P)w\|.$$

(ii) *The minsum locus is nonempty if and only if $Pw \in L$.*

(iii) *Whenever $Pw \in L$, the entire locus is*

$$\text{argmin } F = \bigcup \{[p, q] : p \in A, q \in B, \|p - q\| = \text{dist}(A, B)\}.$$

(iv) *More generally, for $\alpha, \beta > 0$,*

$$\inf_x \{\alpha d(x, A) + \beta d(x, B)\} = \min\{\alpha, \beta\} \text{dist}(A, B),$$

and this weighted infimum is attained if and only if $Pw \in L$. If $\alpha = \beta$, the locus is the union above. If $\alpha < \beta$, it is the set of endpoints in B of nearest pairs; if $\beta < \alpha$, it is the corresponding endpoint set in A .

Proof. For every $x \in \mathcal{H}$, the triangle inequality gives

$$\text{dist}(A, B) \leq d(x, A) + d(x, B).$$

Conversely, evaluating at points of A and then taking the infimum gives

$$\inf_x F(x) \leq \inf_{p \in A} d(p, B) = \text{dist}(A, B),$$

so equality holds.

Suppose first that F has a minimizer x . Closed affine flats are closed convex sets, hence have metric projections. Put $p = P_A x$ and $q = P_B x$. Then

$$\text{dist}(A, B) \leq \|p - q\| \leq \|p - x\| + \|x - q\| = F(x) = \text{dist}(A, B).$$

Thus (p, q) is a nearest pair. Conversely, if (p, q) is a nearest pair and $x = (1 - t)p + tq$, $0 \leq t \leq 1$, then

$$F(x) \leq \|x - p\| + \|x - q\| = \|p - q\| = \text{dist}(A, B),$$

so every point of $[p, q]$ minimizes F . Equality in the Hilbert-space triangle inequality in the previous display forces any minimizer x to lie on the segment joining its projections p and q . This proves both the attainment equivalence and the locus formula.

Differences of points of A and B range over

$$(a - b) + (U + V),$$

because $V = -V$. Hence

$$\text{dist}(A, B) = \text{dist}(w, L) = \text{dist}(w, \bar{L}) = \|(I - P)w\|.$$

The unique nearest point to w in $\bar{L}$ is Pw . Therefore the distance to the algebraic sum L is attained exactly when $Pw \in L$. Together with the nearest-pair equivalence, this proves (i)–(iii).

Finally let $r = d(x, A)$, $s = d(x, B)$, and $d = \text{dist}(A, B)$. Since $r + s \geq d$,

$$\alpha r + \beta s \geq \min\{\alpha, \beta\}(r + s) \geq \min\{\alpha, \beta\}d.$$

The reverse inequality follows by taking points in B when $\alpha \leq \beta$ (or in A when $\beta \leq \alpha$), so that the distance with the larger coefficient vanishes. If the coefficients are unequal, equality also forces the distance to the higher-weight flat to vanish; if they are equal, it forces $r + s = d$. The asserted weighted loci and the same attainment criterion follow. $\square$

4 A positive-distance counterexample

Theorem 2 (Empty locus for two Hilbert affine flats). *There exist two nonempty closed affine flats A, B in a separable real Hilbert space such that*

$$\text{dist}(A, B) = 1, \quad \inf_x (d(x, A) + d(x, B)) = 1, \quad \text{argmin } F = \emptyset.$$

Consequently, Proposition 4.1(c) of the source does not extend to infinite-dimensional Hilbert spaces.

Proof. Let

$$\mathcal{H} = (\ell^2 \oplus \ell^2) \oplus \mathbb{R}, \quad (Dx)_n = \frac{x_n}{n}, \quad y = \left(\frac{1}{n} \right)_{n \geq 1}.$$

The bounded operator $D : \ell^2 \rightarrow \ell^2$ has a closed graph. Moreover, $y \in \ell^2$ but $y \notin \text{ran } D$: a solution of $Dx = y$ would have $x_n = 1$ for every n .

Define

$$A = \ell^2 \oplus \{0\} \oplus \{0\}, \quad B = (0, y, 1) + \{(x, Dx, 0) : x \in \ell^2\}.$$

Both are closed affine flats. Every vector joining a point of A to a point of B has last coordinate one, so $\text{dist}(A, B) \geq 1$.

For $N \geq 1$, let $x^{(N)} = (-1, \dots, -1, 0, 0, \dots)$, with N entries equal to -1 , and put

$$a_N = (x^{(N)}, 0, 0) \in A, \quad b_N = (x^{(N)}, Dx^{(N)} + y, 1) \in B.$$

Then

$$\|a_N - b_N\|^2 = 1 + \sum_{n > N} \frac{1}{n^2} \rightarrow 1.$$

Thus $\text{dist}(A, B) = 1$. If a pair attained this distance, equality with the last-coordinate lower bound would force all other difference coordinates to vanish, in particular $Dx = -y$ for some $x \in \ell^2$. Its only formal solution has $x_n = -1$ for all n , which is not in ℓ^2 . Hence no nearest pair exists. Theorem 1 now gives the stated infimum and the empty locus. $\square$

Corollary 3 (Sharp positive replacement). *For two closed affine flats $a + U$ and $b + V$, closedness of $U + V$ guarantees a nonempty minsum locus for every displacement $b - a$. This applies, in particular, if either direction is finite-dimensional or finite-codimensional. Conversely, every nonclosed sum supplies displacements for which the locus is empty.*

Proof. If $U + V$ is closed, then $P_{\overline{U+V}}(b - a)$ automatically belongs to $U + V$, so Theorem 1 applies. If one direction is finite-dimensional, its image in the quotient by the other direction is a finite-dimensional, hence closed, subspace. If one direction has finite codimension, the same quotient is finite-dimensional. In either case the preimage $U + V$ is closed. For the converse, choose $w \in \overline{U + V} \setminus (U + V)$; then $P_{\overline{U+V}}w = w \notin U + V$. $\square$

The example also embeds in every infinite-dimensional Hilbert space: place it in a closed separable summand and add the orthogonal complement to both affine directions. The two distance functions then depend only on the separable coordinate.

5 A standard positive compactness replacement

For context, reflexivity does recover the source's bounded-target conclusion, even though Theorem 2 shows that it cannot recover the all-affine-flat conclusion.

Proposition 4. *Let X be a reflexive Banach space. Let $K_1, \dots, K_m$ be nonempty norm-closed convex sets, and let γ_i be continuous gauges satisfying $\gamma_i(z) \geq c_i \|z\|$ for constants $c_i > 0$. If one K_i is bounded, then*

$$x \mapsto \sum_{i=1}^m \inf_{y \in K_i} \gamma_i(y - x)$$

attains its minimum, and its minimizer set is bounded and weakly compact.

Proof. Each gauge-distance is convex and norm-continuous, hence weakly lower semicontinuous. If K_j is contained in a norm ball of radius R , then

$$\inf_{y \in K_j} \gamma_j(y - x) \geq c_j \inf_{y \in K_j} \|y - x\| \geq c_j(\|x\| - R).$$

Thus the objective is coercive. A minimizing sequence is bounded, has a weakly convergent subsequence by reflexivity, and weak lower semicontinuity gives a minimizer. The minimizer set is bounded and weakly closed, hence weakly compact. $\square$

6 Verification, limitations, and novelty

The script `code/verify_affine_flat_counterexample.py` checks the explicit formula

$$\|a_N - b_N\| = \sqrt{1 + \sum_{n>N} n^{-2}}$$

for nine truncation sizes through $N = 4096$, confirms convergence to one, and records that the finite-dimensional preimage norm is $\sqrt{N}$ and escapes to infinity. It prints **VERDICT: PASS**. This computation is a sanity check, not part of the proof.

The precise two-flat theorem and counterexample are complete. They do not settle the source’s broader questions for arbitrary finite families, general asymmetric gauges, uniqueness, affine dimension of loci, Hadamard spaces, or manifolds.

A bounded search through 13 August 2026 used the exact source title and arXiv id, source citations, and combinations of the terms *infinite-dimensional minsum*, *Fermat–Torricelli locus affine flats*, *closed affine subspaces distance not attained*, and *nonclosed sum Fermat–Torricelli*. It also checked the 2011 variational-analysis paper arXiv:1009.1594 and the 2023 paper titled *The Generalized Fermat–Torricelli Problem in Hilbert Spaces*. The search found the standard theory of nonclosed subspace sums and later algorithmic Hilbert-space work, but no paper presenting this exact counterexample or criterion as an answer to Section 6. Novelty is therefore plausible, not certified.

Human-review recommendation. Send to a functional analyst. The main checks are the equivalence between minsum attainment and nearest-pair attainment, the membership condition $P_{\overline{U+V}}(b - a) \in U + V$, and the source-to-subproblem match.

References

- [1] T. Jahn, Y. S. Kupitz, H. Martini, and C. Richter, *Minsum Location Extended to Gauges and to Convex Sets*, Journal of Optimization Theory and Applications **166** (2015), 711–746; arXiv:1410.3690; doi:10.1007/s10957-014-0692-6.